

Notes: House, Proebsting, and Tesar (RES, 2026)

Quantifying the Benefits of Labour Mobility in a Currency Union

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1 Big picture: question, objects, and punchline

Question. How valuable is *cross-region labour mobility* as an adjustment mechanism inside a *currency union*? The classic optimum currency area (OCA) logic says that if regions share a currency (no exchange-rate adjustment), then migration can help smooth region-specific shocks. This paper asks: *How big are these benefits quantitatively, and do they depend on what kind of shocks hit the union?*

Setting. The authors compare the euro area (EA) to the United States (U.S.). Empirically:

- unemployment differentials across EA countries are large and persistent relative to U.S. states,
- migration reacts to local labour market conditions in both places, but much more strongly in the U.S.

Core mechanism. In a DSGE model with unemployment, sticky wages/prices, trade, and *endogenous migration*, higher labour mobility makes the local labour supply more elastic: workers move away from depressed regions and toward booming ones. This dampens *unemployment differentials* but changes the inflation–output tradeoff:

- for **demand shocks**, mobility acts like an internal “terms-of-trade” adjustment: it *reduces* inflation and unemployment volatility,
- for **supply shocks** (nontraded productivity), mobility *reduces unemployment* but can *increase* inflation volatility and thus welfare losses.

Main quantitative result. Raising EA mobility to match the U.S. reduces the *welfare cost of a currency union only* when demand-type shocks are an important driver. Under supply-type shocks, higher mobility can raise the cost of the union because it magnifies inflation fluctuations.

2 Empirical section: facts and reduced-form models

2.1 Migration and unemployment dispersion in the EA vs the U.S.

Figure 1 (unemployment rates and dispersion). Unemployment rates vary much more across EA countries than across U.S. states. The cross-sectional standard deviation of unemployment is around **3.6–4.7** in the EA (depending on subperiod) but only about **1.0–1.2** in the U.S. The dispersion spikes during the European sovereign debt crisis.

Figure 2 (gross migration rates). Gross migration rates (flows relative to population) are an order of magnitude larger in the U.S. than in the EA. The EA does see meaningful migration, but the scale is much smaller.

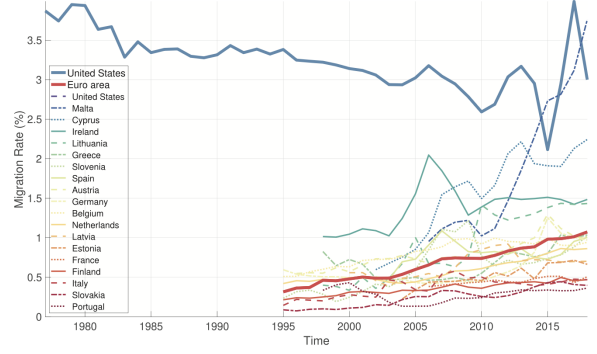
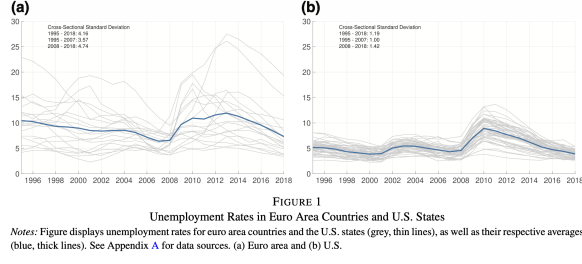


Figure 3 (time series of unemployment dispersion). The cross-sectional standard deviation of unemployment is persistently higher in the EA. This is the “OCA problem”: without exchange rates, region-specific labour market slack can remain for years.

2.2 Empirical objects and equations

2.2.1 Net migration rate

For region i at time t , define the (annual) net migration rate as

$$n_{i,t}^m \equiv \frac{\text{in-migrants}_{i,t} - \text{out-migrants}_{i,t}}{\text{population}_{i,t}}. \quad (3.1)$$

2.2.2 Double-demeaned unemployment

Let $u_{i,t}$ be the unemployment rate. The paper studies region-specific unemployment deviations that net out fixed regional differences and common time shocks:

$$\hat{u}_{i,t}^r \equiv u_{i,t} - \bar{u}_i - (u_t - \bar{u}), \quad (3.2)$$

where $\bar{u}_i$ is region i ’s time average, u_t is the cross-sectional mean in period t , and $\bar{u}$ is the unconditional mean.

2.2.3 Variance decomposition

A useful accounting decomposition is

$$\text{Var}(u_{i,t} - \bar{u}_i) = \text{Var}(\hat{u}_{i,t}^r) + \text{Var}(u_t) + 2 \text{Cov}(\hat{u}_{i,t}^r, u_t). \quad (3.3)$$

Figure 4 (variance decomposition). In the EA, the *idiosyncratic* component $\text{Var}(\hat{u}_{i,t}^r)$ is the dominant share of unemployment variance (about **81%**), while it is much smaller in the U.S. (about

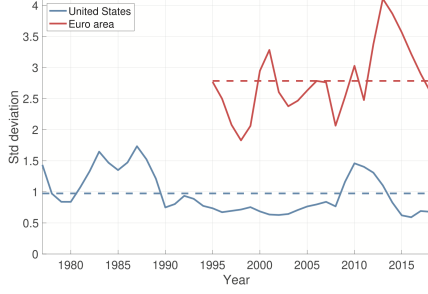


FIGURE 3
Cross-Sectional Standard Deviations in Unemployment Rates

Notes: The figure plots cross-sectional standard deviation in demeaned unemployment rates, $\hat{u}_{i,t}$, for the U.S. states and for the euro area countries. The dotted lines are the respective time averages. See the text for the definition of demeaned unemployment rates. See Appendix A for data sources.

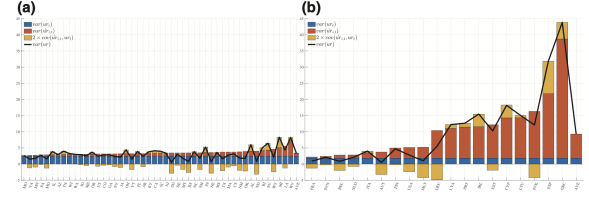


FIGURE 4
Variance Decomposition of the Unemployment Rate

Notes: The figure displays the time-series variance decomposition of the unemployment rates in both the U.S. and the euro area based on equation (3.3). States/Countries are sorted by the idiosyncratic component $\text{var}(\hat{u}_{i,t})$. The figure also displays the composition for the average state/country (AVE) as the bar most to the right. See Appendix A for data sources. (a) U.S. and (b) euro area.

29%). This motivates modelling *country-specific shocks* inside the currency union.

2.2.4 Local projections for persistence

To measure persistence of regional unemployment differentials, the paper uses local projections:

$$\hat{u}_{i,t+h}^r = \beta_0^h \hat{u}_{i,t}^r + \beta_1^h \hat{u}_{i,t-1}^r + \beta_2^h \hat{u}_{i,t-2}^r + \varepsilon_{i,t}^h, \quad h = 0, 1, \dots, H. \quad (3.4)$$

Figure 6 (local projection responses). Regional unemployment shocks decay much faster in the U.S. than in the EA, consistent with stronger internal adjustment mechanisms (migration, wage flexibility, etc.).

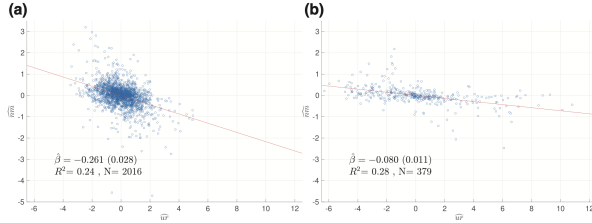


FIGURE 5
Net Migration vs. Unemployment

Notes: The first panel plots the demeaned state net migration rates after controlling for wages, $\hat{m}_{i,t} - \hat{a} \hat{w}_{i,t}$, against the demeaned state unemployment rates $\hat{u}_{i,t}$ for the U.S. over 1977–2018. See equation (3.5). The second panel plots the corresponding data for the sample of euro area countries, 1995–2018. Driscoll and Kraay (1998) standard errors in parentheses. The estimated coefficient on wages is $\hat{a} = 0.007(0.003)$ for the U.S. states and $\hat{a} = 0.002(0.007)$ for the euro area. See Appendix A for data sources. (a) U.S.: 1977–2018 and (b) EURO AREA: 1995–2018.

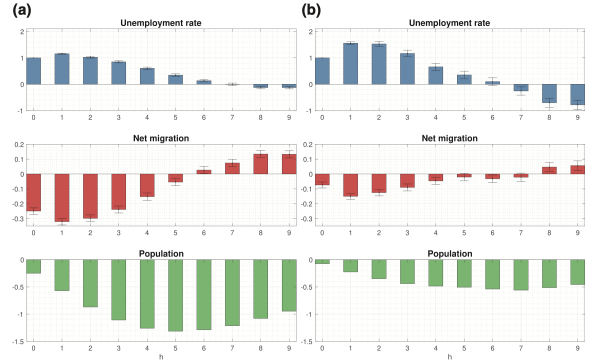


FIGURE 6
Local Projections

Notes: The figure displays the estimated coefficients (and standard errors) from local projection regressions (see equation (3.4)) for the U.S. (panel (a)) and for the euro area (panel (b)). The first set displays the coefficients from regressing the demeaned unemployment rate at time $t+h$, $\hat{u}_{i,t+h}$, on the demeaned unemployment rate at time t , $\hat{u}_{i,t}$, controlling for two lags $\hat{u}_{i,t-1}$ and $\hat{u}_{i,t-2}$. The second set regresses the demeaned net migration rate at time $t+h$, $\hat{m}_{i,t+h}$, on the demeaned unemployment rate at time t , $\hat{u}_{i,t}$, controlling for two lags $\hat{u}_{i,t-1}$ and $\hat{u}_{i,t-2}$. The estimated population response at horizon h is calculated from the estimated coefficients as $(\sum_{k=0}^h (1 + \beta_0^k)) - 1$ of the net migration regression. See Appendix A for data sources. (a) U.S. and (b) euro area.

2.2.5 Migration response to labour market conditions

The key cross-sectional fact is that net migration responds negatively to regional unemployment (controlling for wages):

$$\hat{n}_{i,t}^m = \alpha \hat{w}_{i,t} + \beta \hat{u}_{i,t}^r + \varepsilon_{i,t}. \quad (3.5)$$

Figure 5 (scatter). The estimated unemployment semi-elasticity is much larger in the U.S. than in the EA:

- U.S.: $\hat{\beta} \approx -0.261$ (s.e. 0.028),
- EA: $\hat{\beta} \approx -0.080$ (s.e. 0.011).

Interpretation. When a region’s unemployment is high relative to the union average, people move out in both places, but *about three times more strongly* in the U.S. This empirical gap is what the “high mobility” counterfactual targets in the quantitative model.

3 Model: DSGE in a currency union with endogenous labour mobility

3.1 Environment and agents (high level)

There are multiple countries $i = 1, \dots, N$ inside the euro area. Each country has:

- **capital owners** (immobile) who own capital and firms, choose consumption and investment, and trade bonds;
- **workers** who can migrate across countries, earn labour income when employed, and face unemployment because of nominal wage rigidity;
- **firms** producing differentiated “materials” inputs with sticky prices, and producing traded/nontraded intermediate goods;
- **policy** (taxes, government spending, monetary policy).

Key nominal frictions: Calvo price rigidity for materials prices and Calvo wage rigidity for wages set by unions.

Key real shocks for within-union asymmetries:

- a **preference/“demand” shock** $\omega_{i,t}$ (weight on nontraded goods in final demand),
- a **nontraded productivity/“supply” shock** $Z_{i,t}^N$.

3.2 Capital owners

Lifetime utility of capital owners in country i :

$$V_{i,t}^k = \mathbb{E}_t \sum_{j=0}^{\infty} \beta^j \left[\frac{(c_{i,t+j}^k)^{1-1/\sigma}}{1-1/\sigma} \right]. \quad (4.6)$$

Budget constraint (nominal):

$$P_{i,t}c_{i,t}^k + P_{i,t}I_{i,t} + \frac{B_{i,t}}{(1+i_{e,t})E_{i,t}} - \frac{B_{i,t-1}}{E_{i,t}} = K_{i,t-1} \left(R_{i,t}u_{i,t} - P_{i,t}a(u_{i,t}) \right) + \Pi_{i,t} - T_{i,t}^k. \quad (4.6)$$

Capital accumulation with investment adjustment costs:

$$K_{i,t} = K_{i,t-1}(1 - \delta) + \left[1 - f\left(\frac{I_{i,t}}{I_{i,t-1}}\right) \right] I_{i,t}. \quad (4.6)$$

Notes. $u_{i,t}$ is capital utilization, with cost function $a(\cdot)$; $f(\cdot)$ is an investment adjustment cost. The bond position is $B_{i,t}$, and $E_{i,t}$ is the nominal exchange rate (euro per unit of foreign currency). In a currency union, $E_{i,t}$ is fixed within the union but can move relative to the rest of the world.

3.3 Workers, wages, and unemployment

Workers are hand-to-mouth (consumption equals after-tax labour income):

$$P_{i,t}c_{i,t}^w = (1 - \tau_i^w) W_{i,t} L_{i,t}. \quad (4.7)$$

3.3.1 Labour aggregation and demand

A labour union aggregates differentiated labour types $l \in [0, 1]$ with Stone–Geary preferences:

$$L_{i,t} = \zeta_i + \left[\int_0^1 (L_{i,t}(l) - \zeta_i)^{\frac{\psi_w - 1}{\psi_w}} dl \right]^{\frac{\psi_w}{\psi_w - 1}}. \quad (4.8)$$

Demand for labour type l :

$$L_{i,t}(l) = \zeta_i + \left(\frac{W_{i,t}(l)}{W_{i,t}} \right)^{-\psi_w} (L_{i,t} - \zeta_i). \quad (4.9)$$

The aggregate wage index:

$$W_{i,t} = \left[\int_0^1 (W_{i,t}(l))^{1-\psi_w} dl \right]^{\frac{1}{1-\psi_w}}. \quad (4.9)$$

3.3.2 Calvo wage setting

A fraction $(1 - \theta_w)$ of unions can reset wages each period. The optimal reset wage is

$$(W_{i,t}^*)^{\psi_w} = \frac{\psi_w - 1}{\zeta_i} \frac{\mathbb{E}_t \sum_{j=0}^{\infty} (\theta_w \beta)^j (L_{i,t+j} - \zeta_i) W_{i,t+j}^{\psi_w} P_{i,t+j}^{-1}}{\mathbb{E}_t \sum_{j=0}^{\infty} (\theta_w \beta)^j P_{i,t+j}^{-1}}. \quad (4.10)$$

Wage index evolution:

$$W_{i,t} = \left[\theta_w W_{i,t-1}^{1-\psi_w} + (1-\theta_w) (W_{i,t}^*)^{1-\psi_w} \right]^{\frac{1}{1-\psi_w}}. \quad (4.11)$$

3.3.3 Unemployment

Employment $L_{i,t}$ is determined by firms' labour demand given the wage; labour supply is the worker population $N_{i,t}^w$. Thus unemployment is residual:

$$ur_{i,t} \equiv 1 - \frac{L_{i,t}}{N_{i,t}^w}, \quad \Delta ur_{i,t} \equiv ur_{i,t} - \bar{ur}_i.$$

Linearized wage inflation (wage Phillips curve, as reported in the paper):

$$\pi_{i,t}^w = -\frac{(1-\theta_w\beta)(1-\theta_w)}{\theta_w} \left[\frac{\Delta ur_{i,t}}{1-\bar{ur}_i} \right] + \beta \mathbb{E}_t [\pi_{i,t+1}^w].$$

Interpretation. Higher unemployment pushes wage inflation down because unions are stuck with too-high wages in recessions; with Calvo rigidity, the adjustment is gradual.

3.4 Migration decision and endogenous worker distribution

Each worker chooses a destination j each period to maximize the value of location, with idiosyncratic taste shocks $\varepsilon_{j,t}$ (Type I extreme value). If a worker starts in i , her value is

$$v_{i,t}(\varepsilon_t) = \max_j \left\{ \phi_j U(c_{j,t}^w) + \frac{1}{\gamma} \varepsilon_{j,t} - \tau_j^i + \beta \mathbb{E}_t(V_{j,t+1}) \right\}. \quad (4.12)$$

Integrating out ε_t delivers the log-sum value:

$$V_{i,t} = \frac{1}{\gamma} \ln \left[\sum_j \exp \left(\gamma (\phi_j U(c_{j,t}^w) - \tau_j^i + \beta \mathbb{E}_t(V_{j,t+1})) \right) \right]. \quad (4.13)$$

Migration shares (multinomial logit):

$$n_{j,t}^i = \frac{\exp \left(\gamma (\phi_j U(c_{j,t}^w) - \tau_j^i + \beta \mathbb{E}_t(V_{j,t+1})) \right)}{\sum_k \exp \left(\gamma (\phi_k U(c_{k,t}^w) - \tau_k^i + \beta \mathbb{E}_t(V_{k,t+1})) \right)}. \quad (4.14)$$

Worker population evolves with the migration matrix:

$$N_{i,t}^w = \sum_j n_{i,t}^j N_{j,t-1}^w.$$

A log-linear approximation to bilateral flow changes (from i to j) highlights the key drivers:

$$\begin{aligned} \Delta \ln n_{j,t}^i - \Delta \ln n_{i,t}^i \approx & \gamma \left[\Delta \ln w_{j,t} - \Delta \ln w_{i,t} - \left(\frac{\Delta u r_{j,t}}{1 - \bar{u} r_j} - \frac{\Delta u r_{i,t}}{1 - \bar{u} r_i} \right) \right. \\ & \left. + \beta \mathbb{E}_t (\Delta V_{j,t+1} - \Delta V_{i,t+1}) \right]. \end{aligned} \quad (4.15)$$

Interpretation. Higher real wages and lower unemployment in destination j raise flows toward j . The parameter γ controls *how elastic migration is* to these value differences (high γ means low dispersion of idiosyncratic tastes, so systematic incentives dominate).

3.5 Firms, materials, and nominal price setting

3.5.1 Materials aggregation

Materials $M_{i,t}$ are a CES aggregate of differentiated varieties $m_{i,t}(s)$:

$$M_{i,t} = \left[\int_0^1 m_{i,t}(s)^{\frac{\psi_m - 1}{\psi_m}} ds \right]^{\frac{\psi_m}{\psi_m - 1}}. \quad (4.16)$$

Demand for a variety:

$$m_{i,t}(s) = M_{i,t} \left(\frac{P_{i,t}^m(s)}{P_{i,t}^M} \right)^{-\psi_m}, \quad (4.17)$$

with materials price index

$$P_{i,t}^M = \left[\int_0^1 P_{i,t}^m(s)^{1 - \psi_m} ds \right]^{\frac{1}{1 - \psi_m}}. \quad (4.18)$$

3.5.2 Variety production and factor demands

Each variety producer uses capital and labour:

$$m_{i,t}(s) = K_{i,t}(s)^\alpha L_{i,t}(s)^{1-\alpha}, \quad (4.19)$$

implying a common capital–labour ratio:

$$\frac{K_{i,t}(s)}{L_{i,t}(s)} = \frac{\alpha}{1 - \alpha} \frac{W_{i,t}}{R_{i,t}} = \frac{N_{i,t}^k u_{i,t} K_{i,t-1}}{N_{i,t}^w L_{i,t}}. \quad (4.20)$$

3.5.3 Calvo price setting for materials

Marginal cost (Cobb–Douglas):

$$MC_{i,t} = (W_{i,t})^{1-\alpha} (R_{i,t})^\alpha \left(\frac{1}{1-\alpha} \right)^{1-\alpha} \left(\frac{1}{\alpha} \right)^\alpha.$$

A fraction $(1 - \theta_p)$ can reset prices. The optimal reset price is

$$P_{i,t}^{M,*} = \frac{\psi_m}{\psi_m - 1} \frac{\mathbb{E}_t \sum_{j=0}^{\infty} (\theta_p \beta)^j (c_{i,t+j}^k)^{-1/\sigma} P_{i,t+j}^{-1} (P_{i,t+j}^M)^{\psi_m} MC_{i,t+j} N_{i,t+j} M_{i,t+j}}{\mathbb{E}_t \sum_{j=0}^{\infty} (\theta_p \beta)^j (c_{i,t+j}^k)^{-1/\sigma} P_{i,t+j}^{-1} (P_{i,t+j}^M)^{\psi_m} N_{i,t+j} M_{i,t+j}}. \quad (4.21)$$

Price index evolution:

$$P_{i,t}^M = \left[\theta_p (P_{i,t-1}^M)^{1-\psi_m} + (1 - \theta_p) (P_{i,t}^{M,*})^{1-\psi_m} \right]^{\frac{1}{1-\psi_m}}. \quad (4.22)$$

3.6 Traded and nontraded intermediate goods, and final demand

3.6.1 Traded intermediate good (Eaton–Kortum style)

Traded good demand:

$$Y_{i,t}^T = \left[\int_0^1 (x_{i,t}^T(\nu))^{\frac{\psi_T-1}{\psi_T}} d\nu \right]^{\frac{\psi_T}{\psi_T-1}}. \quad (4.23)$$

The traded goods price index has the Eaton–Kortum form:

$$P_{i,t}^T = \left[\Gamma \left(1 + \frac{1 - \psi_T}{\vartheta} \right) \right]^{\frac{1}{1-\psi_T}} \Phi_{i,t}^{-1/\vartheta}, \quad \Phi_{i,t} = \sum_{j=1}^N Z_j^T (P_{j,t}^M E_{i,t}^j \kappa_i^j)^{-\vartheta}, \quad (4.24)$$

where $E_{i,t}^j \equiv E_{i,t}/E_{j,t}$ and κ_i^j are iceberg trade costs. Import shares are

$$\omega_{i,t}^j = Z_j^T (P_{j,t}^M E_{i,t}^j \kappa_i^j)^{-\vartheta} \Phi_{i,t}^{-1}.$$

3.6.2 Nontraded intermediate good

Nontraded output:

$$Y_{i,t}^N = Z_{i,t}^N M_{i,t}^N, \quad P_{i,t}^N = \frac{P_{i,t}^M}{Z_{i,t}^N}. \quad (4.25)$$

3.6.3 Final good aggregator and demand shocks

Final output combines nontraded and traded intermediates:

$$Y_{i,t} = \left[\omega_{i,t}^{1/\psi_y} (Y_{i,t}^N)^{\frac{\psi_y-1}{\psi_y}} + (1 - \omega_{i,t})^{1/\psi_y} (Y_{i,t}^T)^{\frac{\psi_y-1}{\psi_y}} \right]^{\frac{\psi_y}{\psi_y-1}}. \quad (4.26)$$

Interpretation. $\omega_{i,t}$ is a demand shifter that changes the weight on nontraded goods. This acts like a country-specific demand shock inside the union.

3.7 Monetary policy

In a currency union, the common (euro-area) interest rate follows a Taylor/Mankiw-type rule:

$$i_{e,t} = \phi i_{e,t-1} + (1 - \phi) \left[\bar{r} + \phi_{ur} \sum_{i=1}^N \left(\frac{\bar{N}_i G \bar{D} P_i}{\bar{N}_e G \bar{D} P_e} \right) u r_{i,t} + \phi_{\pi} \sum_{i=1}^N \left(\frac{\bar{N}_i G \bar{D} P_i}{\bar{N}_e G \bar{D} P_e} \right) \pi_{i,t} \right]. \quad (4.27)$$

Interpretation. Policy reacts to *union-wide* unemployment and inflation. Therefore, country-specific shocks inside the union cannot be offset by country-specific interest rate adjustments.

4 Simplified model intuition: why mobility helps for demand shocks but can hurt for supply shocks

Section 5 derives a small open economy approximation that delivers an AD–AS representation.

4.1 Key reduced-form relationship: migration and labour supply

In the simplified setting, the worker population responds to relative labour-market tightness:

$$\tilde{N}_t = \hat{\gamma}(\tilde{w}_t + \tilde{L}_t), \quad \hat{\gamma} = (1 - n^2)\gamma,$$

so a larger migration propensity γ raises the effective labour-supply elasticity.

4.2 Lemma 1: AD and AS

The equilibrium can be written as

$$\tilde{N}_t = -(1 + \vartheta)(\tilde{E}_t + \tilde{P}_t^M) - \Theta_{w,p} \tilde{P}_t^M + \tilde{\omega}_t, \quad (\text{AD})$$

$$\tilde{N}_t = \hat{\gamma}(1 - \omega)(\tilde{E}_t + \tilde{P}_t^M) + \hat{\gamma}[\Theta_p + \Theta_{w,p}] \tilde{P}_t^M + \hat{\gamma}\omega \tilde{Z}_t^N, \quad (\text{AS})$$

where $\Theta_p = \frac{\theta_p}{1 - \theta_p}$ and $\Theta_{w,p} = \frac{\theta_w}{(1 - \theta_p)(1 - \theta_w)}$.

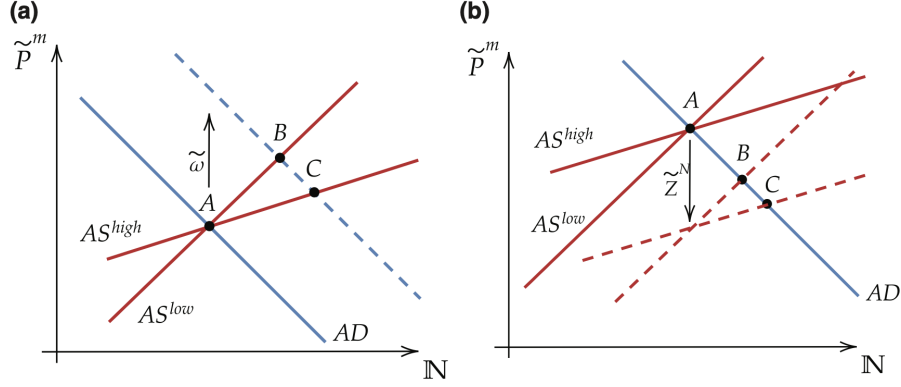


FIGURE 7
Effect of labour mobility in simplified model

Notes: The figure displays the AS and AD curve in the simplified small open economy model with fixed exchange rate (see Lemma 1). Panel A considers an increase in the preference parameter ω , panel B considers an increase in the productivity in the non-traded sector, Z^N . The flatter AS curve corresponds to a model calibration with higher labour mobility (higher γ). (a) Response to positive demand shock and (b) response to positive supply shock.

Figure 7 (graphical intuition). Higher labour mobility corresponds to a higher $\hat{\gamma}$, which *flattens* the AS curve (employment responds more for a given inflation movement). Then:

- **Demand shock** ($\tilde{\omega}_t$): AD shifts. A flatter AS implies *larger employment adjustment and smaller inflation response*. Mobility is stabilizing.
- **Supply shock** ($\tilde{Z}_t^N$): AS shifts. A flatter AS implies *smaller employment drop but larger inflation response*. Mobility stabilizes unemployment but can destabilize inflation.

4.3 Welfare approximation in the simplified model

A second-order approximation (paper's Eq. (5.28)) yields a welfare loss function:

$$\begin{aligned} \Delta W = & -\frac{1}{2} \left[\frac{1}{\sigma} \mathbb{V}(\tilde{C}^w) + \frac{\omega}{1-\omega} \mathbb{V}(\tilde{E} + \tilde{P}^M) + \frac{1-\omega}{\omega} \mathbb{V}(\tilde{P}^M) + \psi_m \Theta_p \mathbb{V}(\tilde{\pi}^M) + (1-\alpha) \Theta_w \mathbb{V}(\tilde{\pi}^w) \right. \\ & \left. + \frac{\sigma}{\sigma-1} \mathbb{E}[\tilde{\omega}_t(\tilde{E}_t + \tilde{P}_t^M)] + \frac{\bar{M}}{\bar{C}} (\bar{u}r - ur^*) + \frac{1}{\gamma} (\bar{N} - \bar{N}^*) + \frac{\beta}{1-\beta} \frac{1}{\gamma} \mathbb{E}[\tilde{N}_t(\tilde{u}_t - \tilde{N}_t)] \right]. \end{aligned} \quad (5.28)$$

How I interpret terms. The first line is variance terms: worker consumption volatility, terms-of-trade volatility, inflation and wage-inflation volatility (nominal rigidity costs). The second line contains mean wedges and migration-option-value terms. Importantly, **inflation volatility is directly welfare-relevant**, so if mobility raises inflation variability under supply shocks, welfare can fall even if unemployment volatility declines.

5 Calibration and estimation: what is pinned down and how

5.1 Table 1: calibrated parameters

Table 1 lists the calibrated parameters and data sources. My quick map:

- preferences: $\beta = 0.99$, $\sigma = 0.5$ (IES), and country-specific steady-state objects (population shares, unemployment rates, nontraded shares),
- nominal rigidities: Calvo wages $\theta_w = 0.86$ and Calvo prices $\theta_p = 0.77$,
- production/trade: capital share $\alpha = 0.30$, depreciation $\delta = 0.02$, trade elasticity $(1 + \vartheta) = 1.5$, etc.,
- monetary policy: persistence $\phi = 0.75$ and response coefficients $\phi_{ur} = -1.40$, $\phi_\pi = 1.40$ (Mankiw-style rule).

Interpretation. This is a quarterly open-economy New Keynesian model with a realistic trade block and with unemployment driven by sticky wages (rather than search/matching).

Parameter	Value	Target/Source
Households, Population and Migration		
Discount factor β	0.99	Standard value
Intertemporal elasticity of substitution σ	0.5	Standard value
Share workers' consumption $\bar{n}_i^w \bar{c}_i^w / (\bar{n}_i \bar{c}_i)$	0.5	Martin and Philippon (2017), Johnson <i>et al.</i> (2006) Fagereng <i>et al.</i> (2021), Jappelli and Pistaferri (2014)
Relative population $\bar{n}_i^w / (\sum_j \bar{n}_j^w)$	c.s.	Eurostat ('95-'15), see Table A10
Location preference parameter ϑ_i	c.s.	$\vartheta_i (c_i^w)^{1-\frac{1}{\sigma}} = 1$
Bilateral migration flows $\bar{n}_{ij}^f$	c.s.	Migration database based on authors' calculation, see Table A12
Nominal wage rigidity θ_w	0.86	Druant <i>et al.</i> (2012)
Unemployment rate $\bar{u}_i$	c.s.	Eurostat ('95-'15), see Table A10
Firms, Production and Trade		
Elasticity of substitution bw. materials ψ_m	10	Basu and Fernald (1995), Basu and Kimball (1997)
Curvature of production function α	0.3	labour income share of 0.63 for Germany Karabounis and Neiman (2013)
Depreciation rate δ	0.02	Annual depreciation rate of 8%
Sticky price probability θ_p	0.77	Price duration: 13 months (Alvarez <i>et al.</i> , 2006)
Trade elasticity $1 + \vartheta$	1.5	e.g. Cravino (2017); Boehm <i>et al.</i> (2023)
Penalty term ι	1e-4	Standard value
Elasticity of substitution bw. y^N and y^T ψ_y	0.44	Stockman and Tesar (1995)
Preference weight non-traded good $\bar{\omega}_i$	c.s.	Share non-traded goods in final demand; OECD TIVA (2005-15), see Table A10
Bilateral trade shares of traded goods $\bar{\omega}_{ij}^t$	c.s.	Import shares for traded goods; OECD TIVA (2005-15), see Table A11
Country size $\bar{n}_i \bar{y}_i / (\sum_j \bar{n}_j \bar{y}_j)$	c.s.	Total demand; OECD TIVA (2005-15), see Table A10
Fiscal and Monetary Policy		
Gov't purchases over final demand $\bar{G}_i / \bar{Y}_i$	c.s.	Eurostat ('95-'15), see Table A10
MP rule persistence ϕ	0.75	Clarida <i>et al.</i> (2000)
MP rule unemployment coefficient ϕ_u	-1.40	Mankiw (2001)
MP rule inflation coefficient ϕ_π	1.40	Mankiw (2001)

Notes: Values marked with c.s. are country- or country-pair specific (see the Appendix for specific values). TIVA: Trade in Value Added Database.

		Model		
Description	Data	Both shocks	Only ω	Only Z^N
Estimated Parameters				
Migration propensity (γ)		0.20	0.20	0.20
Investment adjustment cost (f'')		1.37	1.37	1.37
Utilization adjustment cost (a'')		0.02	0.02	0.02
Persistence preference weights (ρ_w)		0.96	0.96	0.96
Persistence TFP shocks (ρ_Z)		0.88	0.88	0.88
Targeted Moments				
Slope coefficient $\widehat{nm}_{i,t}$ on $\widehat{ur}_{i,t}$	-0.08	-0.08	-0.05	-0.05
Volatility investment to GDP	3.63	3.55	1.20	4.00
Volatility utilization to GDP	0.98	1.02	1.00	1.13
Persistence GDP	0.95	0.94	0.93	0.95
Corr. GDP and unempl.	-0.63	-0.59	-0.88	-0.52
Persistence $e_{i,t}^Z$	0.00	0.09	0.00	0.09
Persistence $e_{i,t}^p$	0.00	0.35	0.35	0.00
Other Moments				
<i>Time-Series Standard Deviation</i>				
Unemployment rate	2.28	2.28	2.41	0.78
Consumption per capita	2.87	2.87	2.02	2.04
Investment per capita	8.05	13.51	2.60	12.53
GDP per capita	2.45	3.86	2.19	3.06
GDP	2.59	3.97	2.22	3.15
Inflation	2.20	2.72	0.62	2.73
Net exports over GDP	1.25	1.01	0.46	1.19
Net migration rate	0.26	0.17	0.15	0.09
<i>Persistence</i>				
Net exports over GDP	0.89	0.97	0.97	0.96
Investment per capita	0.87	0.96	0.97	0.96
Net migration rate	0.65	0.79	0.83	0.74
<i>Correlation with GDP</i>				
Consumption per capita	0.81	0.95	0.93	1.00
Investment per capita	0.84	0.79	0.44	0.99
Net exports over GDP	-0.43	-0.53	0.47	-0.81
Inflation	0.07	0.04	0.85	-0.03

Notes: Moments are calculated based on double-demeaned quarterly data (except moments that refer to net migration, which refer to annual data; in that case, we time-aggregate the simulated quarterly data). The displayed moments are calculated as the weighted average across countries. See Appendices A and C.7.3 for data sources.

5.2 Table 2: estimated parameters and fit

Table 2 estimates five key parameters by matching business-cycle and migration moments:

- migration propensity $\gamma \approx 0.20$,
- investment adjustment cost curvature $f'' \approx 1.37$,
- utilization adjustment cost curvature $a'' \approx 0.02$,
- persistence of demand shock $\rho_\omega \approx 0.96$,
- persistence of nontraded productivity shock $\rho_Z \approx 0.88$.

Moment discipline (high level). The migration parameter γ is disciplined mainly by the empirical slope in Eq. (3.5), i.e. how net migration responds to unemployment differentials. The shock processes are disciplined by persistence/covariances of GDP and unemployment, and by volatilities of investment and utilization.

Fit highlights. With both shocks, the model matches:

- unemployment volatility (targeted) and consumption volatility fairly well,
- the net-migration volatility and persistence moderately well,
- but it overshoots investment volatility (model investment per capita is much more volatile than in data).

The table’s “Only ω ” and “Only Z^N ” columns clarify that neither shock alone can match the joint behaviour of GDP, unemployment, and inflation.

6 Quantitative results

6.1 Table 3: Mundell’s tradeoff quantified

Objects. Table 3 compares macro volatilities and welfare costs under:

- **Fixed** exchange rate (currency union),
- **Float** (independent monetary policy with floating exchange rate),
- **Flex** (flexible prices/wages benchmark),

and under **baseline** EA mobility vs **high** mobility (calibrated to match the U.S. migration–unemployment slope).

Panel A: per-capita stabilization. Under a currency union, higher mobility lowers per-capita unemployment volatility (e.g. from about 2.28 to 2.03). Floating exchange rates reduce unemployment volatility dramatically (to about 0.29) regardless of mobility. Per-capita GDP and consumption volatilities also fall with floating rates, and mobility tends to improve per-capita stabilization under the union.

Panel B: aggregates become more volatile with mobility. With endogenous migration, aggregate GDP and consumption can become *more volatile* when mobility rises, because population flows amplify aggregate fluctuations even if per-capita variables are smoother.

Panel C: inflation. Inflation volatility is high under the union and remains high when mobility increases. This is the key channel through which mobility can become welfare-reducing under supply shocks.

Panel D: welfare costs. The welfare cost of business cycle fluctuations in the union is large (around 1.85% consumption equivalent). With floating rates, the welfare cost is much smaller, but it *increases* when mobility is higher because inflation becomes more volatile under supply shocks.

Panel E: cost of a currency union. Define the cost of the union as welfare loss under **Fixed** minus welfare loss under **Float**. In the baseline calibration, the cost is about 1.31%. With high mobility, it falls to about 1.01% (a reduction of roughly 0.30). But by shock type:

- for **demand shocks only**, higher mobility reduces the union cost (roughly $1.36 \rightarrow 1.02$),
- for **supply shocks only**, higher mobility *raises* the union cost (roughly $0.32 \rightarrow 0.41$).

TABLE 3
Mundell's tradeoff

	Baseline mobility			High mobility	
	(1) Fixed	(2) Float	(3) Flex	(4) Fixed	(5) Float
Panel A: Volatility, Per-Capita Variables (%)					
Unemployment rate	2.28	0.29	0.00	2.03	0.29
GDP per capita	4.68	3.76	3.79	4.23	3.73
Consumption per capita	2.87	2.08	2.11	2.27	1.80
Net migration	0.18	0.16	0.15	0.72	0.64
Panel B: Volatility, Aggregate Variables (%)					
GDP	4.47	3.83	3.90	5.23	5.06
Consumption	3.38	2.84	2.80	4.66	4.33
Panel C: Volatility Measures for Welfare (%)					
Consumption volatility workers	4.60	2.90	3.06	4.02	2.77
Consumption volatility capital owners	3.66	3.26	3.14	5.45	5.05
Wage inflation	0.92	0.37	1.58	0.79	0.40
Material price inflation	0.89	0.50	1.60	0.86	0.56
Panel D: Welfare Costs (% of consumption)					
Consumption volatility workers	0.29	0.13	0.15	0.22	0.11
Consumption volatility capital owners	0.19	0.15	0.14	0.39	0.36
Wage inflation	0.31	0.04	0.00	0.22	0.05
Material price inflation	1.30	0.35	0.00	1.21	0.44
Total	1.85	0.53	0.14	1.74	0.73
Panel E: Cost of Currency Union					
Total	1.31	—	—	1.01	—
Total (Only demand shocks)	1.36	—	—	1.02	—
Total (Only supply shocks)	0.32	—	—	0.41	—

Notes: The table displays several moments as observed in the data (1995–2018) and derived from alternative model simulations. Data moments are based on double-demeaned data. The weighted average across countries is displayed. For the high mobility case, we adjust the migration parameter (γ) to match the slope coefficients for the U.S.. For the floating exchange rate case, all countries follow a Mankiw rule. Inflation measures record annualized rates. The last row displays the cost of the currency union, calculated as the welfare cost under fixed exchange rates less the welfare cost under floating exchange rates.

TABLE 4
Alternative monetary policies

	Both shocks		Only ω shocks		Only Z^N shocks	
	Base	High	Base	High	Base	High
Currency union						
Welfare cost	1.85	1.74	1.75	1.60	0.57	0.82
Mankiw rule						
Welfare cost	0.53	0.73	0.38	0.58	0.25	0.41
Cost of union	1.31	1.01	1.36	1.02	0.32	0.41
Inflation targeting						
Welfare cost	0.22	0.31	0.12	0.23	0.30	0.45
Cost of union	1.62	1.43	1.63	1.37	0.27	0.38
Unemployment targeting						
Welfare cost	0.26	0.34	0.12	0.24	0.29	0.40
Cost of union	1.59	1.40	1.62	1.35	0.28	0.42
Wage inflation targeting						
Welfare cost	0.26	0.33	0.12	0.24	0.29	0.40
Cost of union	1.59	1.40	1.63	1.36	0.28	0.42

Notes: Mankiw rule: $\phi_{ur} = -1.4$ and $\phi_{\pi} = 1.4$. Inflation targeting: $\phi_{ur} = 0$ and $\phi_{\pi} = 1000$. Unemployment targeting: $\phi_{ur} = 100$ and $\phi_{\pi} = 1.4$. Wage inflation targeting: $\phi_{ur} = 0$ and $\phi_w = 1000$.

6.2 Welfare metric in the quantitative model (compensating variation)

The paper reports welfare costs as a consumption-equivalent *compensating variation*. Equation (7.29) expresses the percent change in the aggregate consumption bundle that makes residents of

country i indifferent between (i) the stochastic equilibrium under the benchmark policy and (ii) the stochastic equilibrium under an alternative policy.

Let $\tilde{c}_{i,t}$ denote the *percent deviation* of consumption from the non-stochastic steady state and let $d\mathbb{V}(\cdot)$ denote the change in variance relative to the flexible-price benchmark. Then:

$$\frac{dC_i}{C_i} = \frac{1}{2} \frac{1}{\sigma} \left[\frac{\bar{N}_i^w \bar{c}_i^w}{\bar{N}_i \bar{C}_i} d\mathbb{V}(\tilde{c}_i^w) + \frac{\bar{N}_i^k \bar{c}_i^k}{\bar{N}_i \bar{C}_i} d\mathbb{V}(\tilde{c}_i^k) \right] + \frac{1}{2} \frac{\bar{Y}_i}{\bar{C}_i} [\psi_m \Theta_p d\mathbb{V}(\pi_i) + (1 - \alpha) \Theta_w d\mathbb{V}(\pi_i^w)]. \quad (7.29)$$

Here

$$\Theta_p = \frac{\theta_p}{(1 - \theta_p \beta)(1 - \theta_p)}, \quad \Theta_w = \frac{\theta_w}{(1 - \theta_w \beta)(1 - \theta_w)}.$$

How we read this. Welfare losses come from (i) consumption volatility (for workers and capital owners) and (ii) inefficient price and wage inflation volatility generated by nominal rigidities. Panel D of Table 3 shows that the inflation components are quantitatively dominant (a familiar New Keynesian result), which is why a mechanism that raises inflation volatility (high mobility under supply shocks) can reduce welfare even if it stabilizes unemployment.

6.3 Figure 8: heterogeneity across countries

Figure 8 reports unemployment volatility by country across scenarios. Two notes I take from it:

- the reduction in unemployment volatility from higher mobility is heterogeneous (larger for some peripheral economies),
- floating exchange rates dominate labour mobility as a stabilization margin in the model.

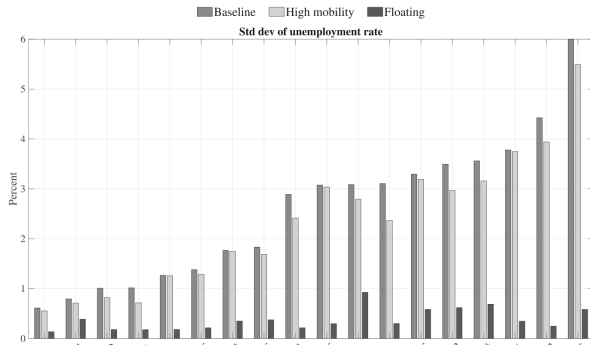


FIGURE 8
Unemployment Volatility by Country

Notes: The figure displays the standard deviation of the unemployment rate in the baseline model, the high-mobility counterfactual and the floating exchange rate counterfactual.

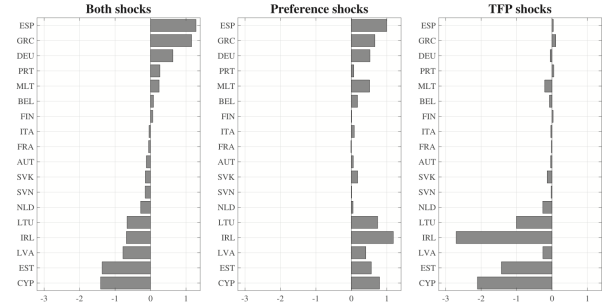


FIGURE 9
Gains from Mobility by Country

Notes: The figure displays the gains from mobility defined as the opportunity cost of the union in the baseline model less the opportunity cost of the union in the high-mobility counterfactual. The opportunity cost of the union is defined as the welfare cost in the union less the welfare cost under floating exchange rates. Results are displayed for the case when both shocks are included, only preference shocks are included or only TFP shocks are included.

6.4 Figure 9 and Table 4: dependence on monetary policy rule and shock mix

Table 4 varies the monetary policy rule under floating exchange rates (Mankiw rule vs strict inflation targeting vs unemployment targeting vs wage-inflation targeting).

Key pattern. Under alternative rules, floating-rate welfare costs can be made very small (strict inflation targeting). Then the implied *cost of a currency union* becomes very large, because the union cannot tailor policy to idiosyncratic shocks.

Role of mobility across rules. In all cases, raising mobility reduces the cost of the union when demand shocks are present, but can raise costs when supply shocks dominate.

Figure 9 (country-level gains). The gains from mobility (reduction in union cost) vary substantially across countries and depend on the shock composition. Countries that are hit relatively more by demand-type disturbances benefit more from migration as an adjustment margin.

7 Conclusion

7.1 Core question and headline answer

The paper asks a quantitative OCA question: *can cross-country labour mobility inside a currency union substitute for independent monetary policy?* The headline answer is *partly, but not uniformly*. Higher labour mobility generally *reduces unemployment fluctuations and lowers the average (population-weighted) opportunity cost of membership in the union*, but it can also *increase aggregate volatility and amplify inflation volatility when supply shocks dominate*. Hence, Mundell-style “mobility substitutes for exchange-rate flexibility” is *state- and country-dependent*.

7.2 Empirical facts that discipline the counterfactual

The empirical motivation is a two-part contrast between the euro area (EA) and the United States (U.S.):

1. **Unemployment dispersion:** unemployment differentials are substantially larger across EA countries than across U.S. states.
2. **Cyclical mobility:** net migration responds to relative unemployment in both regions, but much more strongly in the U.S. A convenient reduced-form summary is the unemployment semi-elasticity of net migration:

$$\beta^{EA} \approx -0.08 \quad \text{vs} \quad \beta^{US} \approx -0.26,$$

i.e., cyclical reallocation is about three times stronger in the U.S.

The main “high-mobility” counterfactual is therefore a disciplined experiment: raise EA migration responsiveness to match the U.S.-estimated sensitivity.

7.3 What the model is used to do

The quantitative engine is a multi-country DSGE model with: (i) sticky prices and wages (New Keynesian transmission), (ii) unemployment from labour-market frictions, (iii) trade linkages across countries, and (iv) **endogenous cross-country migration**. Shocks include **demand-type** disturbances (e.g., preference shifters) and **supply-type** disturbances (e.g., country productivity). The estimation/calibration targets both business-cycle moments and migration moments for the euro area.

The paper then runs two central counterfactuals:

1. **High mobility in the currency union:** raise migration elasticity to U.S. levels while keeping the union (fixed exchange rate and common monetary policy).
2. **Independent monetary policy:** move from the union to floating exchange rates with country-level monetary policy.

Welfare is summarized as a consumption-equivalent cost of business-cycle fluctuations, with an empirically important role for inflation volatility under nominal rigidities.

7.4 Main quantitative conclusions (population-weighted euro-area averages)

The central welfare comparison is between (a) the currency union and (b) floating exchange rates (independent monetary policy). Let the *opportunity cost of the union* be the welfare cost under the union minus the welfare cost under floating exchange rates.

1) Mobility stabilizes unemployment and per-capita quantities, but raises aggregate volatility. Comparing the currency union under baseline mobility to the currency union under high mobility (Table 3, columns (1) vs (4)):

- Unemployment volatility falls from about 2.28 to 2.03 (roughly an 11% reduction).
- Per-capita GDP volatility falls from 4.68 to 4.23; per-capita consumption volatility falls from 2.87 to 2.27.
- However, *aggregate* GDP volatility rises from 4.47 to 5.23, and aggregate consumption volatility rises from 3.38 to 4.66.

Intuition: migration “leans into” regional booms and slumps. Workers move toward booming countries (raising their aggregate output) and away from depressed countries (lowering their aggregate output), so cross-country dispersion of aggregates rises even as per-capita conditions become smoother.

2) Mobility reduces the average opportunity cost of union membership, but does not replicate monetary autonomy. Under baseline mobility, the welfare cost of business cycles is about 1.85% of consumption in the currency union versus 0.53% under floating exchange rates, implying an average opportunity cost of the union of about:

$$1.31\% \quad (\text{baseline mobility}).$$

Under high mobility, the union cost falls to about 1.01%:

$$1.01\% \quad (\text{high mobility}).$$

Thus, **higher mobility reduces the opportunity cost of giving up monetary autonomy** (consistent with Mundell on average), but **independent monetary policy remains quantitatively more powerful** than migration in lowering business-cycle welfare losses.

3) Shock composition is the knife-edge: demand shocks vs supply shocks. A key result is that the welfare sign of “more mobility” depends on whether business cycles are mainly driven by demand-type or supply-type shocks (Table 3, Panel E):

- **Demand shocks only:** the cost of the union falls with mobility (from 1.36 to 1.02).
- **Supply shocks only:** the cost of the union *rises* with mobility (from 0.32 to 0.41).

7.5 Mechanism intuition: why mobility helps for demand shocks

Consider a negative **country-specific demand shock** in a currency union:

1. Labour demand falls and unemployment rises in the shocked country.
2. Out-migration reduces local labour supply, which attenuates unemployment pressure and partially offsets the local contraction.
3. This labour reallocation also tends to *dampen deflationary pressure* in the shocked country, improving the inflation–unemployment tradeoff under sticky prices/wages.

Bottom line: for demand-driven downturns, labour mobility acts like an internal adjustment margin that partially substitutes for region-specific monetary accommodation.

7.6 Mechanism intuition: why mobility can hurt for supply shocks

Now consider a negative **country-specific productivity (supply) shock**:

1. The shock raises marginal costs and creates inflationary pressure (a cost-push component) while activity weakens.

2. Out-migration again reduces unemployment, but it can also tighten local labour supply and reinforce wage/marginal-cost pressure.
3. With nominal rigidities, these interactions can *increase inflation volatility*, which is a first-order contributor to welfare losses in New Keynesian environments.

Bottom line: when supply shocks dominate, mobility can make inflation stabilization harder inside a currency union, increasing the welfare cost of membership even if unemployment volatility falls.

7.7 Cross-country heterogeneity (why some countries gain and others lose)

The average results mask substantial heterogeneity (Figure 9):

- Some large countries (e.g., Germany and Spain) experience sizable welfare gains from higher mobility in the estimated environment.
- Several small/peripheral countries can experience welfare losses under higher mobility in the baseline “both-shocks” environment.
- If only demand shocks are present, *all* countries gain from higher mobility; if only productivity shocks are present, *almost all* countries lose.

Unifying interpretation: a country benefits from higher labour mobility when its cyclical fluctuations are sufficiently demand-driven (mobility stabilizes inflation and unemployment), but it can lose when supply shocks are relatively important (mobility magnifies inflation volatility).

7.8 Policy interpretation

Two policy lessons follow directly from the model-based accounting:

1. **Mobility is not a universal substitute for monetary sovereignty.** It helps most clearly in demand-driven recessions, but it is not guaranteed to improve welfare under supply-driven disturbances.
2. **Inflation stabilization is central.** Alternative policy-rule calculations show that very aggressive inflation stabilization can drive business-cycle welfare costs close to zero under floating exchange rates, underscoring that the key welfare channel is often inflation volatility under nominal rigidities.

7.9 What the conclusion section explicitly flags as “not covered”

The paper isolates labour mobility as one adjustment mechanism. Other potential adjustment margins—trade integration, substitutability between home and foreign goods, financial integration/risk sharing, and coordinated fiscal stabilization—are acknowledged as important but left

outside the scope. A natural next step is to quantify how these margins *interact* with migration, rather than evaluating each in isolation.

7.10 Takeaway

Raising labour mobility inside a currency union generally smooths unemployment and reduces the average cost of giving up monetary independence, but it can increase aggregate volatility and worsen welfare when supply shocks dominate by amplifying inflation volatility; thus, the benefits of mobility are heterogeneous and fundamentally depend on the union's shock composition.